Utility Classes

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Collections: Collections class defines several utility methods for collection objects like sorting, searching, reversing, etc.

Sorting elements of List:

1. public static void sort(List l); ==> Sorts the list in default natural sorting order. The objects present in the list should be homogenous and comparable, otherwise runtime exception will occur saying **ClassCastException**. If the list contains null as object. then runtime exception **NullPointerException** will occur.

2. public static void sort(list l, Comparator c); ==> this method is used for customized sorting order.

```
package Chapter8;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Random;
public class CollectionsSortDemo {
    public static void main(String[] args) {
        ArrayList al = new ArrayList();
        Random r = new Random();
        String alpha = "ABCDEFGHIJKLMNOPRSTUVWXYZ";
        for (int i = 0; i < r.nextInt(10)+5; i++) {
            al.add(alpha.charAt(r.nextInt(26)));
        }
        System.out.println("Before sorting: " + al);
        Collections.sort(al);
        System.out.println("After sorting: " + al);
    }
}
```

Output:

```
Before sorting: [M, S, X, U, W, N, O, C, H, X, H]
After sorting: [C, H, H, M, N, O, S, U, W, X, X]
```

ClassCastException:

```
package Chapter8;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Random;
public class CollectionsSortDemo {
    public static void main(String[] args) {
        ArrayList al = new ArrayList();
        Random r = new Random();
        String alpha = "ABCDEFGHIJKLMNOPRSTUVWXYZ";
```

```
        for (int i = 0; i < r.nextInt(10)+5; i++) {
            al.add(alpha.charAt(r.nextInt(26)));
        }
        al.add(100);
        System.out.println("Before sorting: " + al);
        Collections.sort(al);
        System.out.println("After sorting: " + al);
    }
}
```

Output:

```
Before sorting: [C, C, N, G, V, C, B, K, 100]
Exception in thread "main" java.lang.ClassCastException: class java.lang.Character cannot be cast to class java.lang.Inte
ger (java.lang.Character and java.lang.Integer are in module java.base of loader 'bootstrap')
        at java.base/java.lang.Integer.compareTo(Integer.java:72)
        at java.base/java.util.ComparableTimSort.binarySort(ComparableTimSort.java:262)
        at java.base/java.util.ComparableTimSort.sort(ComparableTimSort.java:189)
        at java.base/java.util.Arrays.sort(Arrays.java:1108)
        at java.base/java.util.Arrays.sort(Arrays.java:1302)
        at java.base/java.util.ArrayList.sort(ArrayList.java:1804)
        at java.base/java.util.Collections.sort(Collections.java:145)
        at Chapter8.CollectionsSortDemo.main(CollectionsSortDemo.java:17)
```

NullPointerException

```
package Chapter8;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Random;
public class CollectionsSortDemo {
    public static void main(String[] args) {
        ArrayList al = new ArrayList();
        Random r = new Random();
        String alpha = "ABCDEFGHIJKLMNOPRSTUVWXYZ";
        for (int i = 0; i < r.nextInt(10)+5; i++) {
            al.add(alpha.charAt(r.nextInt(26)));
        }
        //al.add(100);
        al.add(null);
        System.out.println("Before sorting: " + al);
        Collections.sort(al);
        System.out.println("After sorting: " + al);
    }
}
```

Output:

```
Before sorting: [B, J, E, C, O, S, N, V, F, O, null]
Exception in thread "main" java.lang.NullPointerException: Cannot invoke "java.lang.Comparable.compareTo(Object)" because
 "pivot" is null
        at java.base/java.util.ComparableTimSort.binarySort(ComparableTimSort.java:262)
        at java.base/java.util.ComparableTimSort.sort(ComparableTimSort.java:189)
        at java.base/java.util.Arrays.sort(Arrays.java:1108)
        at java.base/java.util.Arrays.sort(Arrays.java:1302)
        at java.base/java.util.ArrayList.sort(ArrayList.java:1804)
        at java.base/java.util.Collections.sort(Collections.java:145)
        at Chapter8.CollectionsSortDemo.main(CollectionsSortDemo.java:18)
```

Customized Sorting order

```java
package Chapter8;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Random;

public class CollectionsSortDemoCSO {
   public static void main(String[] args) {
      ArrayList al = new ArrayList();
      Random r = new Random();
      String alpha = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
      for (int i = 0; i < r.nextInt(10)+5; i++) {
         al.add(alpha.charAt(r.nextInt(26)));
      }
      System.out.println("Before sorting: " + al);
      Collections.sort(al, new MyComparator());
      System.out.println("After sorting: " + al);
   }
}
```

```java
package Chapter8;
import java.util.Comparator;
public class MyComparator implements Comparator{
   @Override
   public int compare(Object o1, Object o2) {
      Character c1 = (Character)o1;
      Character c2 = (Character)o2;
      return c2.compareTo(c1);
   }

}
```

Output:

```
Before sorting: [V, A, Q, Z, E, D, O, B]
After sorting: [Z, V, Q, O, E, D, B, A]
```